# Desmos Graphing Tactics, Regression & Systems Strategy

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

**1** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the value of $f(2)$?

**Answer:** $-6$. Read the graph at $x = 2$: the line passes through $(2, -6)$, so $f(2) = -6$.

**2** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

What is the positive $x$-intercept of the graph of $y = x^2 + 3x - 4$ in the $xy$-plane?

**Answer:** $1$. Setting $y = 0$: $(x - 1)(x + 4) = 0$, so $x = 1$ or $x = -4$. The positive intercept is 1.

**3** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

What is the positive $x$-intercept of the graph of $y = x^2 + x - 6$ in the $xy$-plane?

**Answer:** $2$. Setting $y = 0$: $(x - 2)(x + 3) = 0$, so $x = 2$ or $x = -3$. The positive intercept is 2.

**4** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = x$ and $y = -x - 2$ intersect at the point $(a, b)$ in the $xy$-plane. What is the value of $a$?

**Answer:** $-1$. Set the expressions equal: $x = -x - 2$, giving $2x = -2$ and $a = -1$.

**5** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = 4x - 2$ and $y = -2x + 16$ intersect at the point $(a, b)$ in the $xy$-plane. What is the value of $a$?

**Answer:** $3$. Set the expressions equal: $4x - 2 = -2x + 16$, giving $6x = 18$ and $a = 3$.

**6** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = x + 3$ and $y = -3x + 7$ intersect at the point $(a, b)$ in the $xy$-plane. What is the value of $a$?

**Answer:** $1$. Set the expressions equal: $x + 3 = -3x + 7$, giving $4x = 4$ and $a = 1$.

**7** [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = 4x$ and $y = -3x - 21$ intersect at the point $(a, b)$ in the $xy$-plane. What is the value of $a$?

**Answer:** $-3$. Set the expressions equal: $4x = -3x - 21$, giving $7x = -21$ and $a = -3$.

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## Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

**8** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $1$. The vertex of the parabola shown is $(0, 1)$, so $h = 0$, $k = 1$, and $h + k = 1$.

**9** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $-4$. The vertex of the parabola shown is $(-2, -2)$, so $h = -2$, $k = -2$, and $h + k = -4$.

**10** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $-2$. The vertex of the parabola shown is $(0, -2)$, so $h = 0$, $k = -2$, and $h + k = -2$.

**11** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $-2$. The vertex of the parabola shown is $(1, -3)$, so $h = 1$, $k = -3$, and $h + k = -2$.

**12** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $-1$. The vertex of the parabola shown is $(3, -4)$, so $h = 3$, $k = -4$, and $h + k = -1$.

**13** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**Answer:** $0$. The vertex of the parabola shown is $(2, -2)$, so $h = 2$, $k = -2$, and $h + k = 0$.

**14** [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation of a circle in the $xy$-plane is $(x + 5)^2 + (y - 2)^2 = 9$. What is the radius of the circle?

**Answer:** $3$. In $(x - h)^2 + (y - k)^2 = r^2$, $r^2 = 9$, so $r = 3$.

## Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

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**15** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the $xy$-plane, the line $y = 4x - c$ and the parabola $y = x^2 - 3$ intersect at exactly one point, where $c$ is a constant. What is the value of $c$?

**Answer:** 7. Set $x^2 - 3 = 4x - c$: $x^2 - 4x + (c - 3) = 0$. One intersection means the discriminant is 0: $16 - 4(c - 3) = 0$, so $c = 7$.

**16** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 8x + 6y - 39 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**Answer:** 8. Completing the square: $(x - 4)^2 + (y + 3)^2 = 64$, so the radius is 8.

**17** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 4)(x - 3)$ intersects the $x$-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

**Answer:** 7. The $x$-intercepts are -4 and 3, so $q - p = 3 - (-4) = 7$.

**18** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 10x - 2y - 38 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**Answer:** 8. Completing the square: $(x + 5)^2 + (y - 1)^2 = 64$, so the radius is 8.

**19** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 4x + 2y - 59 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**Answer:** 8. Completing the square: $(x + 2)^2 + (y + 1)^2 = 64$, so the radius is 8.

**20** [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 8x + 10y - 23 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**Answer:** 8. Completing the square: $(x - 4)^2 + (y + 5)^2 = 64$, so the radius is 8.

## Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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**21** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the value of $f(2)$?

**a.** 3

**b.** 2

**c.** $-7$

**d.** 7 ✓

Read the graph at $x = 2$: the line passes through $(2, 7)$, so $f(2) = 7$.

**22** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the value of $f(2)$?

**a.** $-1$

**b.** 5

**c.** $-5$ ✓

**d.** $-3$

Read the graph at $x = 2$: the line passes through $(2, -5)$, so $f(2) = -5$.

**23** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the value of $f(2)$?

**a.** 2

**b.** $-4$

**c.** 4 ✓

**d.** 0

Read the graph at $x = 2$: the line passes through $(2, 4)$, so $f(2) = 4$.

**24** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the value of $f(2)$?

**a.** 2

**b.** 1

**c.** $-4$

**d.** 4 ✓

Read the graph at $x = 2$: the line passes through $(2, 4)$, so $f(2) = 4$.

**25** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the slope of the graph of $y = f(x)$?

**a.** 5

**b.** 3 ✓

**c.** $-3$

**d.** 4

The line passes through $(0, -3)$ and $(2, 3)$, so its slope is $\frac{3-(-3)}{2-0} = 3$.

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**26** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the slope of the graph of $y = f(x)$?

**a.** 2 ✓ **b.** $-2$

**c.** 3 **d.** 4

The line passes through $(0, -2)$ and $(2, 2)$, so its slope is $\frac{2-(-2)}{2-0} = 2$.

**27** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the slope of the graph of $y = f(x)$?

**a.** 2 **b.** $-2$ ✓

**c.** 4 **d.** 0

The line passes through $(0, 4)$ and $(2, 0)$, so its slope is $\frac{0-(4)}{2-0} = -2$.

**28** [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function $f$ is shown. What is the slope of the graph of $y = f(x)$?

**a.** 3 **b.** $-2$

**c.** 2 ✓ **d.** $-1$

The line passes through $(0, -1)$ and $(2, 3)$, so its slope is $\frac{3-(-1)}{2-0} = 2$.

## Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

**29** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**a.** 4 **b.** 2 ✓

**c.** $-2$ **d.** 0

The vertex of the parabola shown is $(0, 2)$, so $h = 0$, $k = 2$, and $h + k = 2$.

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**30** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**a.** 1

**b.** $-1$

**c.** 2

**d.** $-3$ ✓

The vertex of the parabola shown is $(-1, -2)$, so $h = -1$, $k = -2$, and $h + k = -3$.

**31** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x - h)^2 + k$, where $h$ and $k$ are constants, is shown. What is the value of $h + k$?

**a.** $-2$

**b.** 1 ✓

**c.** $-3$

**d.** 3

The vertex of the parabola shown is $(-1, 2)$, so $h = -1$, $k = 2$, and $h + k = 1$.

**32** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 9$ and $y = 4x$ have in the $xy$-plane?

**a.** 3

**b.** 0 ✓

**c.** 1

**d.** 2

Setting $x^2 + 9 = 4x$ gives $x^2 - 4x + 9 = 0$ with discriminant $16 - 36 = -20$, which determines the number of intersections.

**33** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 2$ and $y = 2x$ have in the $xy$-plane?

**a.** 0 ✓

**b.** 1

**c.** 2

**d.** 3

Setting $x^2 + 2 = 2x$ gives $x^2 - 2x + 2 = 0$ with discriminant $4 - 8 = -4$, which determines the number of intersections.

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**34** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation of a circle in the $xy$-plane is $(x + 1)^2 + (y + 2)^2 = 16$. What is the radius of the circle?

**a.** 4 ✓

**b.** 16

**c.** 8

**d.** 5

In $(x - h)^2 + (y - k)^2 = r^2$, $r^2 = 16$, so $r = 4$.

**35** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 1$ and $y = 3x$ have in the $xy$-plane?

**a.** 1

**b.** 3

**c.** 2 ✓

**d.** 0

Setting $x^2 + 1 = 3x$ gives $x^2 - 3x + 1 = 0$ with discriminant $9 - 4 = 5$, which determines the number of intersections.

**36** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation of a circle in the $xy$-plane is $(x - 4)^2 + (y - 2)^2 = 16$. What is the radius of the circle?

**a.** 4 ✓

**b.** 16

**c.** 8

**d.** 5

In $(x - h)^2 + (y - k)^2 = r^2$, $r^2 = 16$, so $r = 4$.

**37** [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 7$ and $y = 2x$ have in the $xy$-plane?

**a.** 0 ✓

**b.** 1

**c.** 2

**d.** 3

Setting $x^2 + 7 = 2x$ gives $x^2 - 2x + 7 = 0$ with discriminant $4 - 28 = -24$, which determines the number of intersections.

## Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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**38** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the $xy$-plane, the line $y = 2x - c$ and the parabola $y = x^2 + 4$ intersect at exactly one point, where $c$ is a constant. What is the value of $c$?

**a.** 4

**b.** $-4$

**c.** $-3$ ✓

**d.** 3

Set $x^2 + 4 = 2x - c$: $x^2 - 2x + (c + 4) = 0$. One intersection means the discriminant is 0: $4 - 4(c + 4) = 0$, so $c = -3$.

**39** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 4)(x - 2)$ intersects the $x$-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

**a.** 2

**b.** 6 ✓

**c.** $-2$

**d.** $-8$

The $x$-intercepts are -4 and 2, so $q - p = 2 - (-4) = 6$.

**40** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 4x - 6y - 23 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**a.** 8

**b.** 6 ✓

**c.** 36

**d.** 23

Completing the square: $(x + 2)^2 + (y - 3)^2 = 36$, so the radius is 6.

**41** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 2)(x - 1)$ intersects the $x$-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

**a.** 3 ✓

**b.** $-1$

**c.** $-2$

**d.** 1

The $x$-intercepts are -2 and 1, so $q - p = 1 - (-2) = 3$.

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**42** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 6x + 8y - 39 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**a.** 64

**b.** 39

**c.** 10

**d.** 8 ✓

Completing the square: $(x - 3)^2 + (y + 4)^2 = 64$, so the radius is 8.

**43** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the $xy$-plane, the line $y = 4x - c$ and the parabola $y = x^2 - 4$ intersect at exactly one point, where $c$ is a constant. What is the value of $c$?

**a.** $-8$

**b.** 16

**c.** 4

**d.** 8 ✓

Set $x^2 - 4 = 4x - c$: $x^2 - 4x + (c - 4) = 0$. One intersection means the discriminant is 0: $16 - 4(c - 4) = 0$, so $c = 8$.

**44** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = -(x)(x - 1)$ intersects the $x$-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

**a.** 0

**b.** 3

**c.** $-3$

**d.** 1 ✓

The $x$-intercepts are 0 and 1, so $q - p = 1 - (0) = 1$.

**45** [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 2x - 3 = 0$ defines a circle in the $xy$-plane. What is the radius of the circle?

**a.** 4

**b.** 3

**c.** $-2$

**d.** 2 ✓

Completing the square: $(x - 1)^2 + (y)^2 = 4$, so the radius is 2.

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